

# EUGENE OBUNDE OKUMU

Software Engineer | React · Python / FastAPI · PostgreSQL · Docker · API Integration

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## PROFILE

Full-stack software engineer who designs, builds, tests and deploys data-driven web platforms end to end: relational database design, secure REST APIs, third-party API integration and containerised React/Next.js front ends. Co-built Flowguard, a multi-tenant data platform for critical pipeline infrastructure, taking it from schema design to a Dockerised production deployment. Experienced in moving features from development through testing into production, automating end-to-end and API tests, and working closely with non-technical stakeholders to turn requirements into usable tools. Motivated to apply these skills to environmental data and digital public goods that support the 2030 Agenda.

## TECHNICAL SKILLS

- **Frontend:** React, Next.js, Vue, JavaScript, Tailwind CSS, HTML, CSS; component-based, responsive UI design
- **Backend & APIs:** Python, FastAPI, Java, Spring Boot, Django, Ruby on Rails; REST API design and third-party API integration
- **Databases & Data:** PostgreSQL, SQL, Firebase (NoSQL), SQLAlchemy, Alembic migrations, schema design, data cleaning and transformation, pandas
- **Containerisation & Deployment:** Docker, Docker Compose, Render, Git/GitHub; development-to-production releases
- **Testing & Quality:** Playwright end-to-end testing, API testing (curl), unit testing, QA on production systems
- **Security:** JWT authentication, bcrypt, role-based access control, multi-tenant data isolation
- **Data & AI:** XGBoost, feature engineering, prompt engineering
- **Ways of working:** Agile Scrum, sprint planning, stakeholder communication, team leadership, cross-functional collaboration

## EXPERIENCE

**Full-Stack Engineer (Apprentice)** — E&M Tech House, Inuka Fellowship Program Jul – Sep 2026

- Designed and built backend services and REST APIs in FastAPI and Spring Boot, integrated with Next.js front ends on live client projects.
- Delivered features end to end, from database integration and API endpoints to the user-facing views that consume them, and moved them into production.
- Served as QA on production products (dealer finance, Boda Plus, CRM), automating end-to-end tests with Playwright and API checks with curl to keep releases stable.

**Frontend Developer (Industrial Attachment)** — Web3Clubs Jun – Aug 2025

- Built user interfaces for decentralised applications using React, Vue, Tailwind CSS and JavaScript.
- Integrated front ends with external services, working with blockchain engineers to connect Solidity smart contracts to the UI.

**Team Lead (Attachment)** — JHub, JKUAT May – Aug 2024

- Led the team that built Mavuno Market, a digital marketplace connecting farmers directly with buyers, from concept to delivery using Agile practices.
- Owned product vision and UI design, ran sprint coordination and translated stakeholder needs into features.

## PROJECTS

**Flowguard: Multi-Tenant Data Platform for Pipeline Infrastructure** *FastAPI, PostgreSQL, React/Next.js, Docker*

- Co-built (team of five) a platform that helps fluid-transport operators predict pump failure risk 7 days ahead and act before breakdowns, using Kenya Pipeline Company's 1,342 km network as the case study.
- Designed the modular backend architecture and PostgreSQL schema, enforcing tenant isolation in every table and query so multiple organisations can share one deployment safely.
- Built the data pipeline that cleans operational data and engineers features for XGBoost models, plus an alert engine driven by per-organisation thresholds.
- Implemented JWT/bcrypt authentication, role-based access control and invite-based onboarding with guards against privilege escalation.
- Containerised the full stack with Docker and deployed it to production on Render.

**Mavuno Market: Farmer-to-Buyer Marketplace** *JHub, JKUAT*

- Led delivery of a digital marketplace for smallholder farmers, from concept and UI design to a working product.

**Aminia DApp: Supply-Chain Tracking** *React.js, Web3*

- Built the React front end for a supply-chain tracking application, integrating Web3 wallet and smart-contract calls.

## EDUCATION

**BSc, Computer Technology** — Jomo Kenyatta University of Agriculture & Technology · Second Class Honours, Upper Division 2026

**KCSE** — Maranda High School · B+ 2018 – 2021

## CERTIFICATIONS, TRAINING & LANGUAGES

- **Aviva Technologies Ltd** — Intensive Web Engineering Program (10 months)
- **PLP Africa** — Data Analytics and AI (in progress)
- **Microsoft Learn** — 31 badges, including AI Fundamentals and Prompt Engineering
- **JHub, JKUAT** — IP Fundamentals, Cybersecurity Basics, Personal Branding
- **Languages:** English (fluent, spoken and written) · Kiswahili (fluent, spoken and written)